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AADK TASK 2

Image Inclusion & Resource Management in Compose

MindMatrix

1. What is the Image Composable?

The Image composable in Jetpack Compose is used to display images on the screen. It allows us to show pictures from drawable resources, URLs, or other image sources inside our app UI.

We use it along with a painter (like painterResource) to load an image from the drawable folder.

Example use-case:
Displaying a profile picture at the top of a business card app screen.

Example idea:

- A profile image at the top
- Name and details shown below using Text

2. Why Image Density Matters in Android

Android devices have different screen sizes and pixel densities (low, medium, high, etc.). Because of this, images must be managed properly for different screen types.

Folders like:

- drawable-mdpi
- drawable-hdpi
- drawable-xhdpi
- drawable-nodpi

help Android choose the correct image size for each device.

If we ignore image density:

- The image may look blurry on high-resolution screens.
- It may appear too large or too small.
- The app performance may decrease if very large images are used.
- It can increase app size unnecessarily.

Using drawable-nodpi is helpful when we don't want Android to scale the image automatically.

3. Two Best Practices When Adding Images

1) Use Proper Naming Conventions

Image file names should:

- Be lowercase
- Not contain spaces
- Use underscores instead of spaces
- Example: profile_image.png

This avoids errors in Android Studio.

2) Optimize Image Size

Images should not be too large in size (MB).

We should:

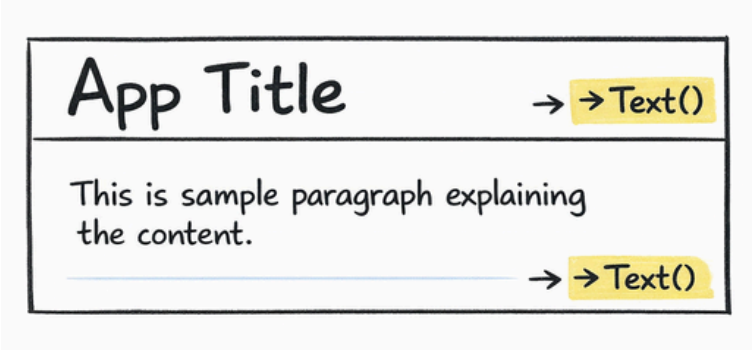
- Compress images before adding them.
- Use appropriate resolution.
- Avoid adding very high-resolution images if not required.

This improves:

- App performance
- Loading speed
- Memory usage

4. Visual Storyboard

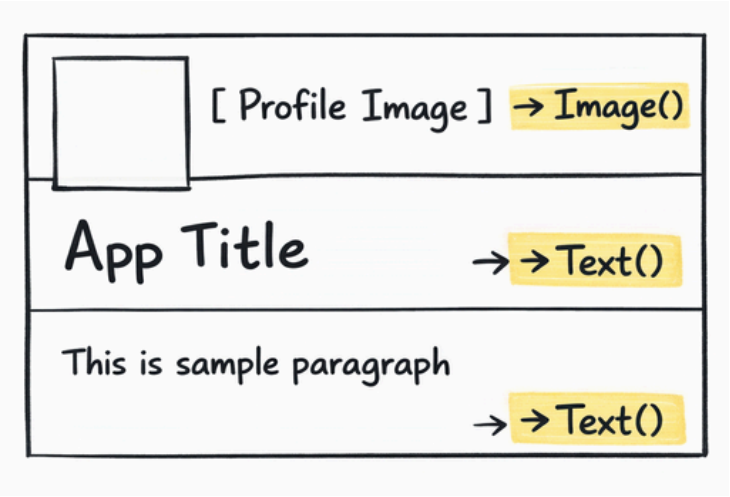
Frame 1 — Placeholder Text Only



Composables Used:

- Column (main layout)
- Text (title and paragraph)

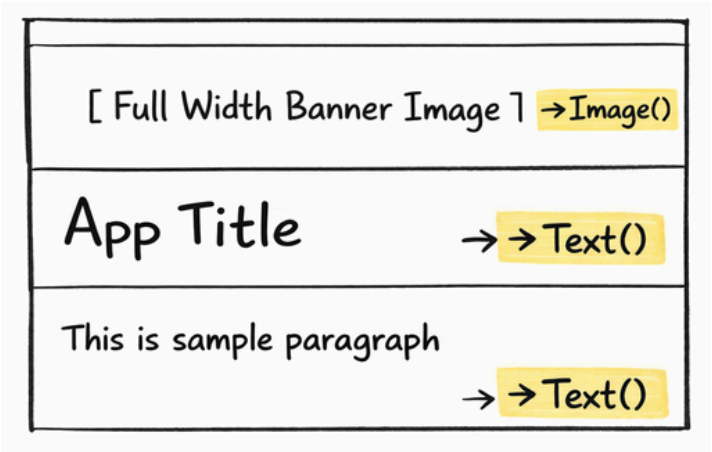
Frame 2 — Image Added at the Top



Composables Used:

- Column (vertical arrangement)
- Image (at top)
- Text (below image)

Frame 3 — Image Full Width



In this frame:

- The image is set to fill maximum width.
- We may use Modifier.fillMaxWidth()
- ContentScale.Crop can be used to crop the image properly.

Composables Used:

- Column
- Image (fillMaxWidth + crop)
- Text

This shows how image placement changes the overall layout appearance.

5. Reflection — Challenges Images Bring to UI Design

Images can create challenges like increasing app size and reducing performance if they are not optimized properly. Large images can slow down loading time. Also, layout alignment becomes difficult if image size is not handled correctly. Accessibility is another issue because images need proper content descriptions for screen readers. So we must plan image usage carefully before adding it to the app.